

Vavuniya University
Faculty of Applied Science
Department of Physical Science
Information and Communication Technology
Second Semester

IT 3243(P) & CSH3254(P) – Parallel Computing (Practical)
In Course Assessment – 01

Time Allowed: **40 minutes**

Date: **09. 04. 2026**

Answer all questions within allocated time.

1. Write an MPI program to compute the similarity between two integer sequences using the **Cosine Similarity** measure.

$$\text{Cosine Similarity} = \frac{\sum(A_i \times B_i)}{\sqrt{\sum A_i^2} \times \sqrt{\sum B_i^2}}$$

Consider this example for calculating Cosine Similarity

A = {1, 2, 3}

B = {4, 5, 6}

$$\text{Cosine Similarity} = \frac{(1 \times 4 + 2 \times 5 + 3 \times 6)}{\sqrt{1^2 + 2^2 + 3^2} \times \sqrt{4^2 + 5^2 + 6^2}}$$

Initially, the process with **Id 0** has the following sequences:

Sequence – A: {12, 25, 37, 44, 59, 63, 71, 82, 91, 10, 22, 35, 48, 50, 67, 73, 85, 90, 14, 18, 21, 27, 33, 39, 46, 52, 60, 66, 75, 88, 95, 99}

Sequence – B: {11, 20, 35, 40, 55, 60, 70, 80, 90, 15, 25, 30, 45, 55, 65, 70, 80, 85, 10, 20, 22, 28, 30, 38, 48, 50, 62, 68, 72, 85, 90, 100}

- a. Master should distribute **Sequence A** among **four processes**. Each process should receive **8 integer values** from Sequence A.
- b. Master should distribute **Sequence B** among **four processes**. Each process should receive **8 integer values** from Sequence B.
- c. Each process, including process with Id 0, should print the received portion of **Sequence B**.
- d. Each process should compute and print its **partial dot product**:

$$\sum(A_i \times B_i)$$

for its assigned portion.

- e. Each process should also compute and print:
 - i. Partial sum of squares of A $\rightarrow \sum A_i^2$
 - ii. Partial sum of squares of B $\rightarrow \sum B_i^2$
- f. Process with id 0 should collect all partial results from other processes and compute:
 - i. Total dot product
 - ii. Total sum of squares for A and B
- g. Finally, process with Id 0 should compute and print **Cosine Similarity**.